

which has to chunter along with bits that are only on or off.

Quantum computation, a tool for science, represents a fundamental shift that is now underway. Quantum supremacy was the name given to the hypothetical point at which a quantum computer could perform a calculation that no conceivable digital computer could perform in a reasonable amount of time. From a scientific point of view, the future of quantum computation is now much more exciting. The outputs of early digital computers could be verified by hand calculations, while the outputs of quantum computers have until now been verifiable by digital computers.

Quantum teleportation is based on quantum entanglement, through which the properties of a quantum particle can be transferred to a distant particle without physical movement of the particle itself. It is just two interlinked particles revealing the properties of the other particles. Quantum teleportation is possible by using photons which carry the quantum

information about the two possible states in case of qubits.

Building quantum computers takes phenomenal engineering. They must be isolated to ensure nothing interferes with the delicate quantum states of the qubits. This is why they are kept in vacuum chambers containing fewer particles than outer space, or in refrigerators colder than anything in the universe. Scientists have been able to teleport qubits.

One of the greatest challenges is controlling or removing quantum decoherence. This usually means isolating the system from its environment as interactions with the external world cause the system to decohere. However, other sources of decoherence also exist. Examples include the quantum gates, and the lattice vibrations and background thermonuclear spin of the physical system used to implement the qubits. Decoherence is irreversible.

Splitting the photon into three beams through the use of an intricate contraption consisting of the calibrated setup of lasers, beam splitters and

barium borate crystals, the researchers created qutrits. They also said that it could be possible to use ququarts in the future. The researchers were able to prove that qutrit teleportation is possible. Despite rapid and impressive experimental progress, most researchers believe that “fault-tolerant quantum computing [is] still a rather distant dream”.

No scalable quantum computing hardware had been demonstrated till September 2019. Nevertheless, there is an increasing amount of investment in quantum computing by governments, established companies, and startups. Current research focuses on building and using near-term intermediate-scale devices and demonstrating quantum supremacy alongside the long-term goal of building and using a powerful and error-free quantum computer.

Quantum based computers will arrive in the market sooner than we think. What is most exciting is not what we can do with a quantum computer today, but the undiscovered truths it will reveal tomorrow! **EFY**

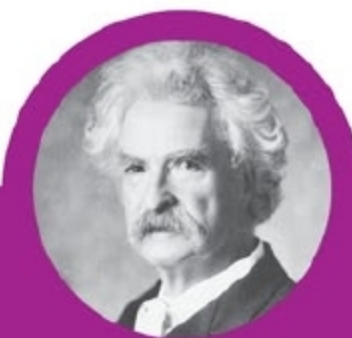
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Many a small thing has been made large by the right kind of advertising.

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-Mark Twain

